F78PA Probability & Statistics A

Tutorial 8

1. The joint probability density function of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} \frac{6}{7}\left(x^2 + \frac{xy}{2}\right), & 0 \le x \le 1, 0 \le y \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Sketch the region in the plane where $f_{X,Y}(x,y) > 0$.

   (b) Verify that the function $f_{X,Y}$ *is* a density function.

   (c) Find the marginal density functions f_X and f_Y for X and Y, respectively, and determine $\mathbb{P}\{1/4 < X < 3/4\}$ and $\mathbb{P}\{1/2 < Y\}$.
   Are the variables X and Y independent? Give reasons for your answer.

   (d) Compute $\mathbb{P}(X > 2Y)$. (Note: You need to set up a double integral to do this.)

   (e) Compute the conditional probability $\mathbb{P}(Y > 1 | X < 1/2)$.

2. Suppose that the random variables X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} ae^{-y/2}, & 0 \le x \le y < \infty, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Sketch the region in the plane where $f_{X,Y}(x,y) > 0$ and determine the constant a.

   (b) Find the marginal densities f_X and f_Y for X and Y, respectively, and identify the distributions of these two random variables.
   Are the variables X and Y independent? Give reasons for your answer.

3. The joint density function of X and Y is given by

$$f_{X,Y}(x,y) = \begin{cases} x\exp(-x(y+1)), & x \ge 0, y \ge 0, \\ 0, & \text{otherwise.} \end{cases}$$

   (a) Find the marginal densities f_X and f_Y for X and Y, and determine $\mathbb{P}\{X > 1\}$ and $\mathbb{P}\{1 \le Y \le 2\}$.

   (b) Find $f_{X|Y}(x|y)$, the conditional density of X, given $Y = y$, and $f_{Y|X}(y|x)$, the conditional density of Y, given that $X = x$.
   Use these conditional densities to determine the conditional probabilities $\mathbb{P}\{X > 1 \mid Y = 1\}$ and $\mathbb{P}\{1 \le Y \le 3 \mid X = 2\}$

   (c) Find the density function $f_Z(z)$ for $Z = XY$.
   Hint: For $z > 0$, find $F_Z(z) = \mathbb{P}(Z \le z) = \mathbb{P}(XY \le z)$ first. You will have to set up a double integral to do this.

4. Suppose that the random variables X and Y have joint density

$$f_{X,Y}(x,y) = \begin{cases} ae^{-\lambda x - \mu y}, & x \ge 0, \quad y \ge 0, \\ 0, & \text{otherwise,} \end{cases}$$

   where λ, μ are positive constants.

   (a) Find the constant a.

   (b) Find the marginal densities f_X and f_Y of X and Y.
   Are X and Y independent? Give reasons for your answer.

   (c) Find $\mathbb{P}(X < Y)$.

5. Suppose that X is a continuous random variable with $\mathrm{Im}X \subseteq [0, \infty)$ and with density function $f_X(x)$.

   (a) Show that
   $$E(X) = \int_0^\infty \mathbb{P}\{X > t\}dt.$$
   Hint: Use the fact that
   $$\int_{t=0}^\infty \mathbb{P}\{X > t\}dt = \int_{t=0}^\infty \left(\int_{x=t}^\infty f_X(x)dx \right) dt.$$

   (b) Suppose that $X \sim Exp(\lambda)$ where $\lambda > 0$. Use part (a) to show that $E(X) = 1/\lambda$.

Exam Style Question

Let X and Y be jointly continuous random variables with density function

$$f_{X,Y}(x, y) = \begin{cases} 7x^5 & \text{if } 0 < y < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Show that $f_{X,Y}(x, y)$ is a valid joint density function. (2 marks)

(b) Find the marginal density functions $f_X(x)$ and $f_Y(y)$. (4 marks)

(c) Are X and Y independent? Justify your answer. (1 mark)

(d) Determine the expected values $\mathbb{E}[X]$ and $E[Y]$. (2 marks)

(e) Let $Z = Y/X$. Find the cumulative distribution function F_Z of Z. (6 marks)